

Minimal Communication Protocols for Swarm Coordination Under Bandwidth Constraints

A quantitative study of information budgets, topology, and emergent communication

Tier 2 - Master's Research Design, revised as the post-capstone extension

Decentralized Swarm Emergence Research Program

Stage One revision - September 2026

Evidence status: Stage One validates the experimental environment, instrumentation, reward accounting, heuristic baselines, and reproducibility pipeline. It does not yet establish learned emergence, a critical alpha threshold, learned communication, or a phase transition.

Abstract

After establishing the no-communication reward-topology baseline, this extension treats communication as a measurable resource rather than a binary feature. Agents operate under the same decentralized execution constraints while receiving a bounded message budget b . The study asks how much information exchange is required to improve coordination, whether the benefit saturates, and how communication topology changes the performance-cost frontier. No bandwidth threshold is claimed in advance; b^* is an estimand that exists only if the data support a stable, reproducible improvement over the no-communication baseline.

1. Dependency on the capstone

This tier begins only after the learned alpha experiment is frozen. The capstone provides a validated environment, no-communication reference policies, reward-topology controls, and behavioral metrics. Communication is then introduced as a new independent variable rather than being mixed into the original capstone matrix.

2. Research question

What communication budget and topology are sufficient for decentralized agents to obtain meaningful coordination gains relative to matched no-communication policies?

3. Communication model

At time t , agent i observes local state $o_{i,t}$, emits bounded message $m_{i,t}$, receives messages permitted by topology G_t , and chooses action $a_{i,t}$. The experimental communication budget b may be implemented through message dimensionality, quantization, transmission frequency, or a controlled combination. Per-agent and system-wide communication costs are reported separately so that scaling does not hide aggregate information use.

4. Experimental factors

Factor	Core levels / treatment
Communication budget b	0 plus low-to-high bounded sweep
Topology	broadcast; local neighborhood; graph-based
Protocol	fixed/control; learned
Reward topology	selected alpha regimes from capstone
Perturbation	message corruption/dropout ablation
Training	matched budgets and seed policy

5. Primary outcomes

- Task success and completion efficiency.

- Communication cost per successful episode and per completed target.
- Marginal performance gain per additional unit of communication.
- Mutual information between messages and actions, and between messages and task-relevant state, when estimable.
- Message redundancy, temporal consistency, and clustering structure.
- Generalization under shifted layouts and communication perturbations.

6. Threshold discipline

A critical bandwidth b^* will be reported only if a pre-specified improvement criterion is met across independent training runs and held-out evaluation. A smooth monotonic curve, immediate gain at the smallest nonzero budget, saturation without a sharp threshold, or no communication benefit are all admissible results. Phase-transition language will remain descriptive unless the empirical evidence warrants a stronger formal interpretation.

7. Signature analysis

The central figure is a bandwidth-coordination frontier: coordination performance on the vertical axis and communication cost on the horizontal axis, stratified by topology. Secondary plots show marginal information gain, robustness to message corruption, and whether learned messages encode task-relevant structure.

8. Contribution boundary

This tier does not assume that learned messages are symbolic language, that communication is universally beneficial, or that simulated bandwidth maps directly onto a particular radio/network stack. It studies information-constrained coordination in the implemented MARL laboratory.

Figures

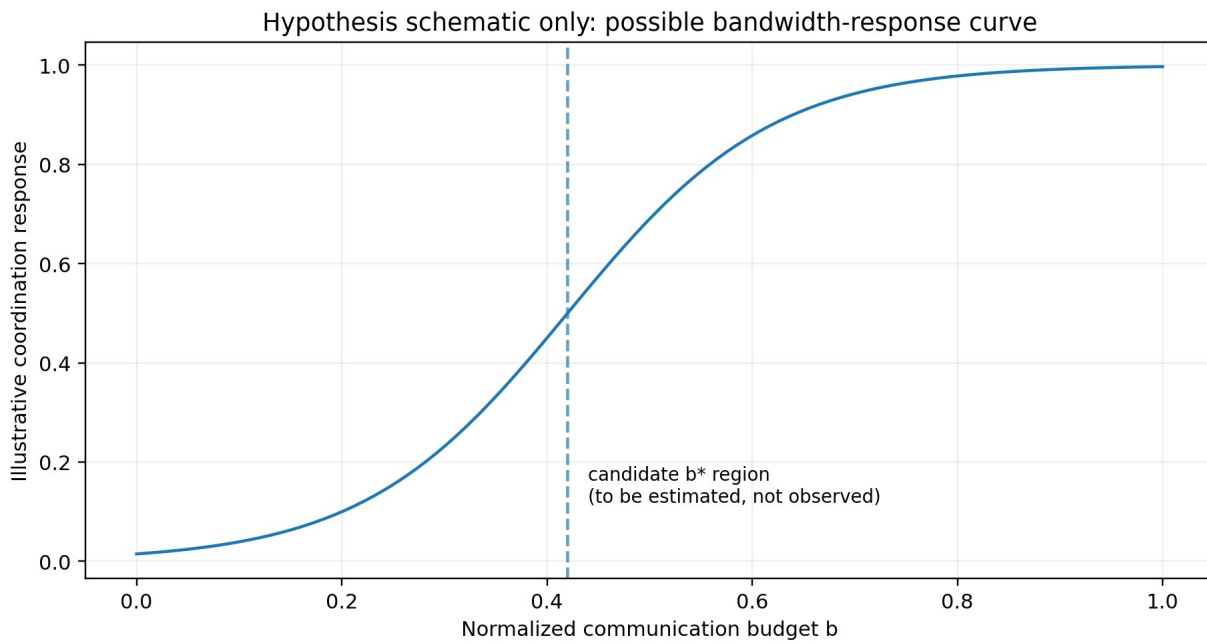


Figure 1. Hypothesis schematic only. The curve and candidate b^ region are illustrative and must not be interpreted as measured results.*